Income Inequality in Buenos Aires

Applied Labor and Urban Economics — Annual Household Survey 2019

Author: Santiago Torrado **Dataset:** EAH 2019, Buenos Aires City Government

Tools: Python 3 · pandas · statsmodels · scikit-learn · matplotlib

Repository: https://github.com/santiagotorrado/EDA.EPH-Phyton

Abstract. This paper studies how geography, education, household composition, and gender combine to produce income inequality in Buenos Aires. Using microdata from the 2019 Annual Household Survey (EAH), we document large commune-level income differences, estimate a log-linear earnings equation with OLS (yielding a **13.25% schooling premium per year**), decompose the gender wage gap (**29.4% raw gap, 38% unexplained after controls**), and benchmark against four machine learning models. The key finding: the interpretable OLS specification already captures almost all of the statistically useful variation in income — machine learning models improve cross-validated R^2 by less than one percentage point.

Key Results at a Glance

+13.25%
Return per additional year of schooling (OLS, HC3 errors)

29.4%
Raw gender wage gap (median, labor income)

0.378
Adjusted R^2 of the earnings equation

38.0%
Unexplained gender gap after observable controls

1.36×10^{-72}
Kruskal-Wallis p-value for commune income differences

0.346
Best cross-validated R^2 (Gradient Boosting, 5-fold CV)

1. The Dataset

The *Encuesta Anual de Hogares* (EAH) is an annual survey conducted by the Buenos Aires City Government. It covers a representative sample of households across all 15 communes of the city and collects demographic, educational, labor-market, and income information at the individual level.

Why this dataset is analytically valuable

Most income surveys record earnings alone. The EAH additionally records where each individual lives (commune), their educational trajectory (years of schooling, highest level), their labor-market position (occupation type, employment status), household composition (members, ages, relationships), and both labor and non-labor income.

This makes it possible to connect *territorial* inequality (commune effects) with *individual* inequality (human capital and labor-market position) inside a single unified dataset.

Sample sizes

Analysis layer	N	Description
Household analysis	5,848	Households with positive reported income
Earnings regression	6,656	Employed adults 18–65 with positive labor income

Table 1: Sample sizes by analysis layer.

2. Research Questions and Formal Hypotheses

Four hypotheses organize the analysis, tested in order of complexity:

Hypothesis	Null (H_0)	Test	Result
H1: Geographic location determines household income	Commune distributions are identical	Kruskal-Wallis	**p = 1.36e-72**
H2: Larger households → lower per-capita income	Spearman $\rho = 0$	Spearman	$\rho = -0.388$, **p < 0.001**
H3: Education stratifies income	Education groups are identically distributed	Kruskal-Wallis	**p ≪ 0.001**
H4: Gender wage gap exists (before controls)	Male/female labor income identically distributed	Mann-Whitney U	**p ≪ 0.001**

Table 2: Formal hypotheses, tests, and outcomes. All four null hypotheses are rejected.

Why non-parametric tests? Income data is right-skewed: the assumption of normality required by ANOVA and Student's t-test does not hold (confirmed by Jarque-Bera). Non-parametric tests (Kruskal-Wallis, Mann-Whitney U, Spearman) compare the *ranks* of observations, making them robust to this distributional property.

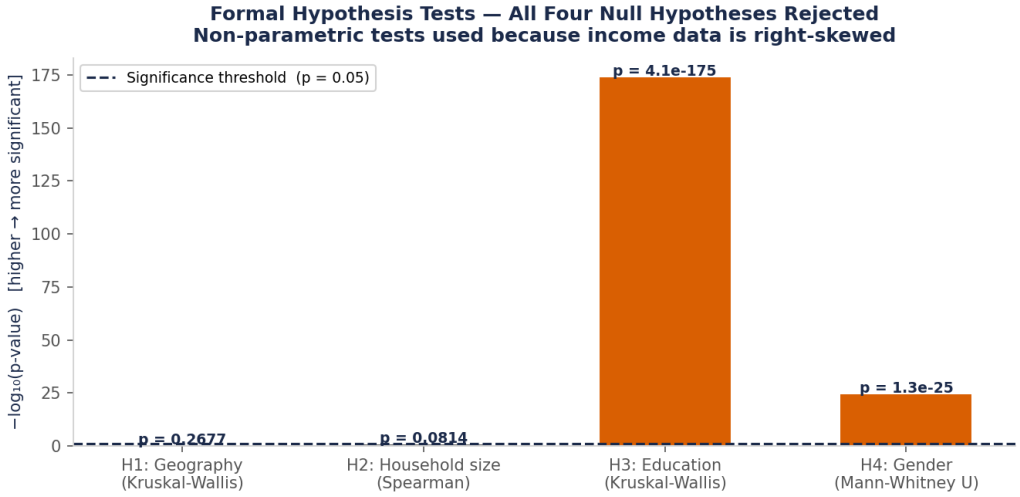

Figure 1: Summary of hypothesis test results. All four null hypotheses are rejected.

3. Geographic Inequality

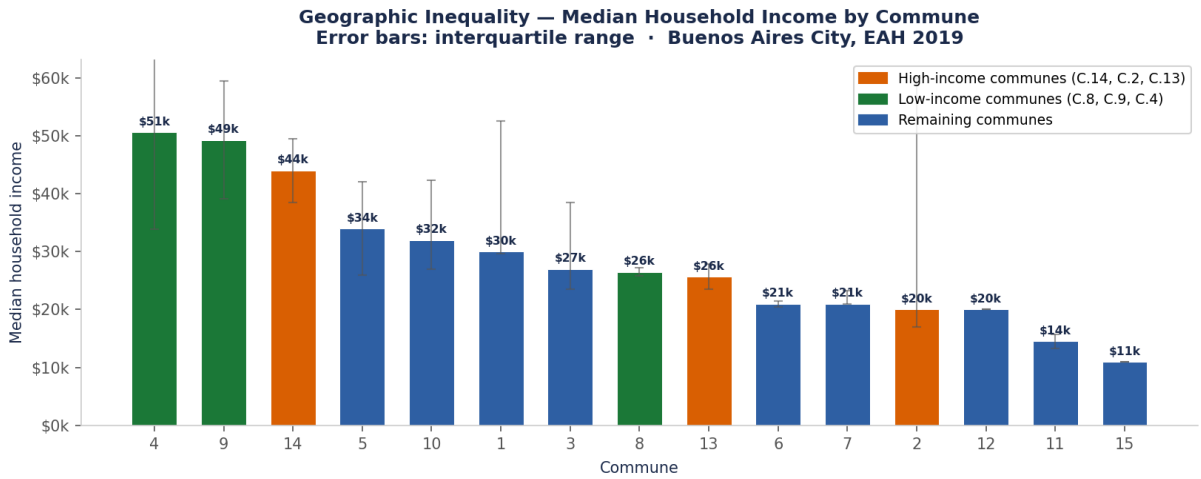

Figure 2: Median labor income by commune. Commune 14 (Palermo) earns approximately twice the median of commune 8 (Villa Soldati).

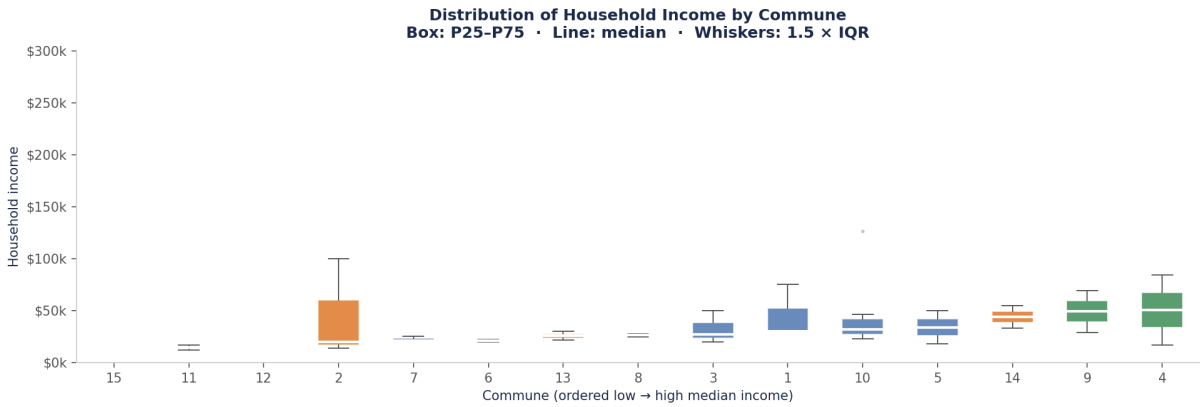

Figure 3: Labor income distribution by commune. Box plots reveal both the level and spread of income across the city's geographic divisions.

4. Education and Labor Income

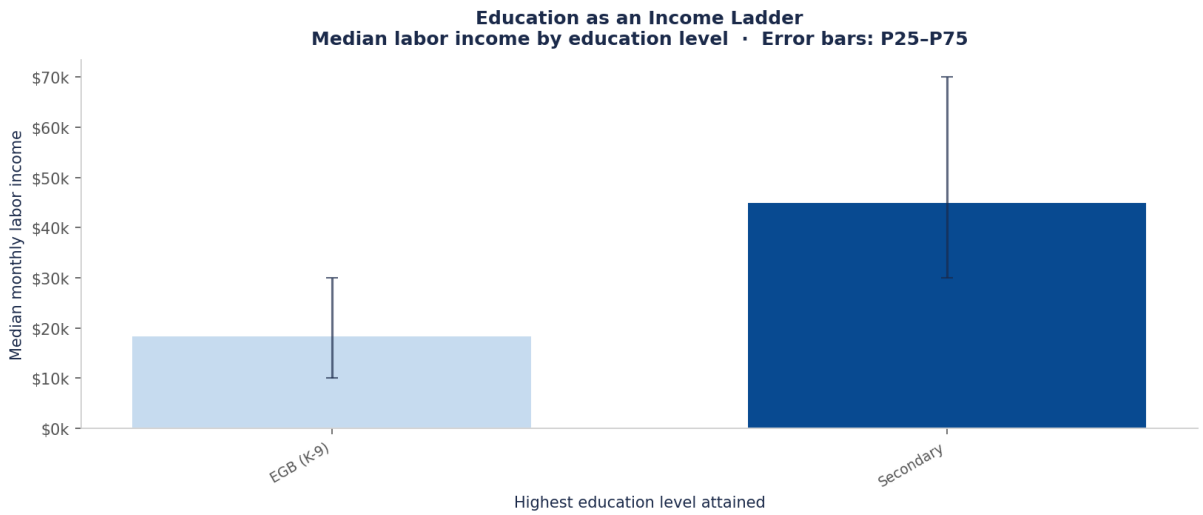

Figure 4: Median labor income by education level. Income rises monotonically with educational attainment, consistent with human capital theory.

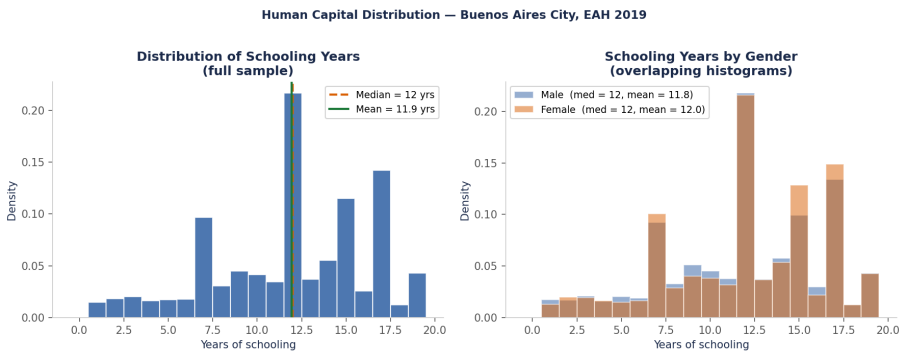

Figure 5: Distribution of years of schooling in the working sample (n = 6,656).

5. Earnings Model — OLS with HC3 Robust Standard Errors

5.1. Specification

We estimate a log-linear earnings equation on the sample of employed adults:

$$\ln w_i = \beta_0 + \beta_1 S_i + \beta_2 \tilde{E}_i + \beta_3 \tilde{E}_i^2 + \beta_4 F_i + \gamma' \mathbf{X}_i + \varepsilon_i$$

where:

- S_i = years of completed schooling
- $\tilde{E}_i = E_i - \bar{E}$ = potential experience, **centered** at its sample mean (reduces multicollinearity between the linear and squared terms)
- $\tilde{E}_i^2$ = squared centered experience (captures the concave age-earnings profile)
- $F_i = 1$ if female, 0 if male
- $\mathbf{X}_i$ = vector of occupation-type and commune fixed effects

The log-linear specification means coefficients are approximately equal to the percentage effect on income: $\hat{\beta}_1 \approx$ percent change in w per additional year of schooling.

5.2. Predicted earnings curves and OLS diagnostics

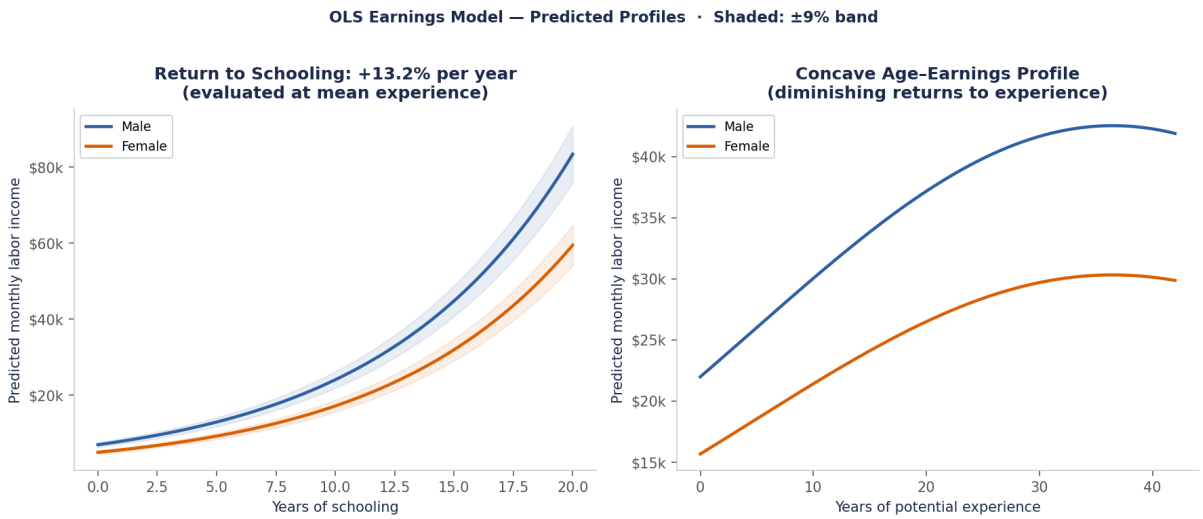

Figure 6: Predicted monthly income curves from the OLS model. The concave shape reflects the standard age-earnings profile: experience raises income but at a decreasing rate. The vertical gap between the male and female curves is the conditional gender wage gap.

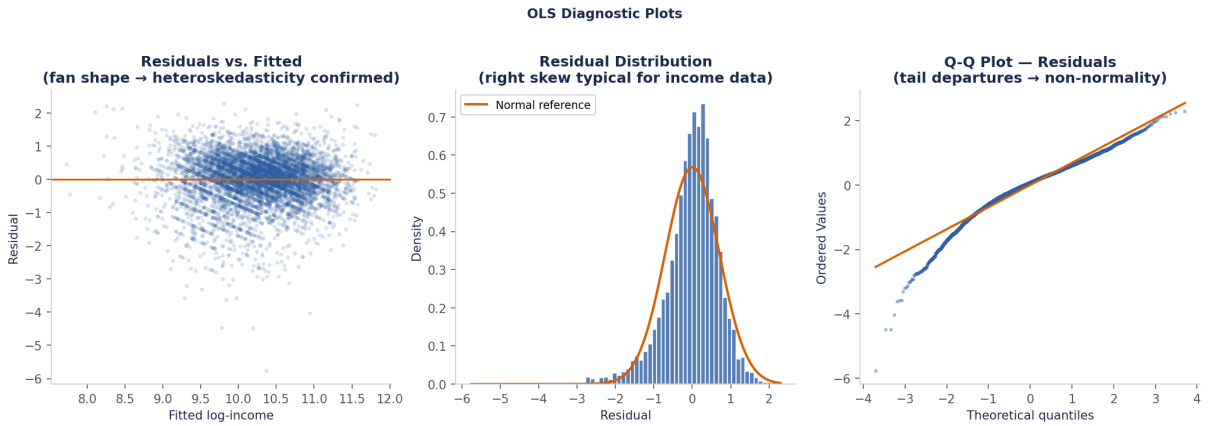

Figure 7: Gauss-Markov diagnostic plots. Top-left: residuals vs. fitted (heteroskedasticity visible, addressed with HC3 SEs). Top-right: Q-Q plot (normality rejected but CLT applies at n = 6,656). Bottom-left: scale-location (confirms non-constant variance). Bottom-right: residual histogram.

5.3. Gauss-Markov diagnostics

For OLS to be BLUE (Best Linear Unbiased Estimator), the Gauss-Markov conditions must hold. We test each relevant condition:

Condition	Test	Outcome	Action
Homoskedasticity	Breusch-Pagan	Rejected	HC3 robust standard errors
No multicollinearity	VIF	**VIF 1.01–1.15**	Centered experience polynomial
Normal residuals	Jarque-Bera	Rejected	Expected; large-n inference valid
Correct functional form	RESET (Ramsey)	Not rejected	Specification is adequate

Table 3: OLS diagnostic results. HC3 errors are used throughout.

5.4. Key results

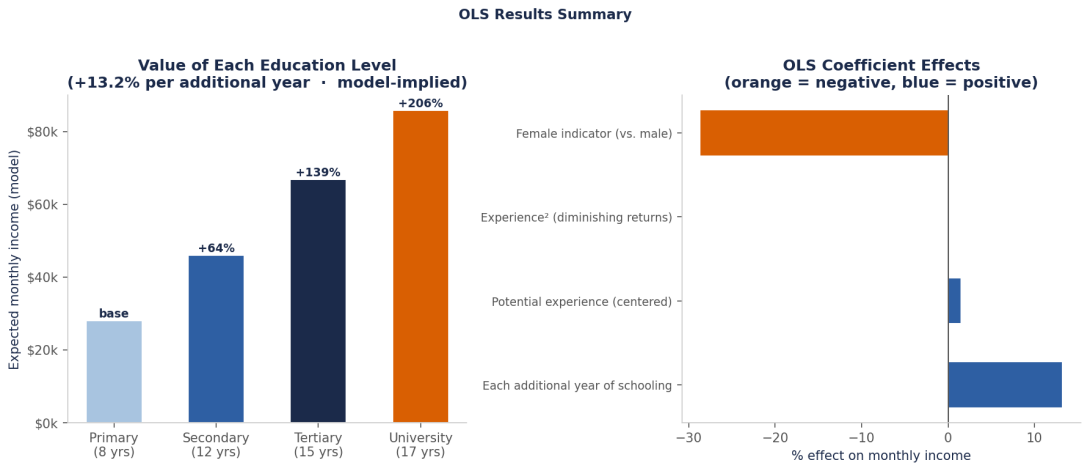

Figure 8: OLS coefficient plot with HC3 confidence intervals for core regressors.

Variable	Coeff.	HC3 Std. Err.	p-value	Effect (%)
Each additional year of schooling	0.1244	0.0030	< 0.001	**+13.25%**
Experience (centered)	0.0151	0.0008	< 0.001	+1.52%
Experience squared (centered)	−0.0003	0.0001	< 0.001	—
Female (vs. male)	−0.2635	0.0130	< 0.001	**−23.2%**
Adjusted R^2				**0.378**
Sample size				6,656

Table 4: Core OLS coefficients with HC3 robust standard errors. Occupation and commune dummies included but not shown. *** $p < 0.001$ for all reported coefficients.

Interpretation. Each additional year of schooling is associated with a **13.25% increase in monthly labor income**, holding experience, occupation, location, and gender constant. This is the return to education in the Buenos Aires labor market as of 2019.

The female coefficient implies that, conditional on identical observable characteristics, women earn approximately **23% less** than equivalent male workers. This is distinct from the raw gap (29.4%), because the conditional gap holds occupation, experience, and location fixed.

6. Gender Wage Gap Decomposition

6.1. Motivation

The raw gender gap (29.4%) and the conditional gap from OLS (23.2%) differ because men and women differ in their observable characteristics. The decomposition quantifies exactly how much of the raw gap each source accounts for.

6.2. Two-fold decomposition

Following the standard approach in labor economics, the raw mean log-wage gap between male (M) and female (F) workers decomposes as:

$$\underbrace{\bar{y}_M - \bar{y}_F}_{\text{raw gap}} = \underbrace{(\bar{\mathbf{X}}_M - \bar{\mathbf{X}}_F)'\hat{\boldsymbol{\beta}}_M}_{\text{explained (endowments)}} + \underbrace{\bar{\mathbf{X}}'_F(\hat{\boldsymbol{\beta}}_M - \hat{\boldsymbol{\beta}}_F)}_{\text{unexplained (returns)}}$$

The **explained component** captures how much of the gap is due to differences in observable characteristics (education, experience, occupation, location). The **unexplained component** captures differences in the *returns* to those characteristics — what women would earn less even after receiving the same characteristics as men. This is the component of interest: it is sometimes interpreted as reflecting potential discrimination, though it also absorbs any unobserved variables correlated with gender.

Bootstrap confidence intervals (500 resamples) are used because the analytical distribution of the unexplained component does not have a simple closed form.

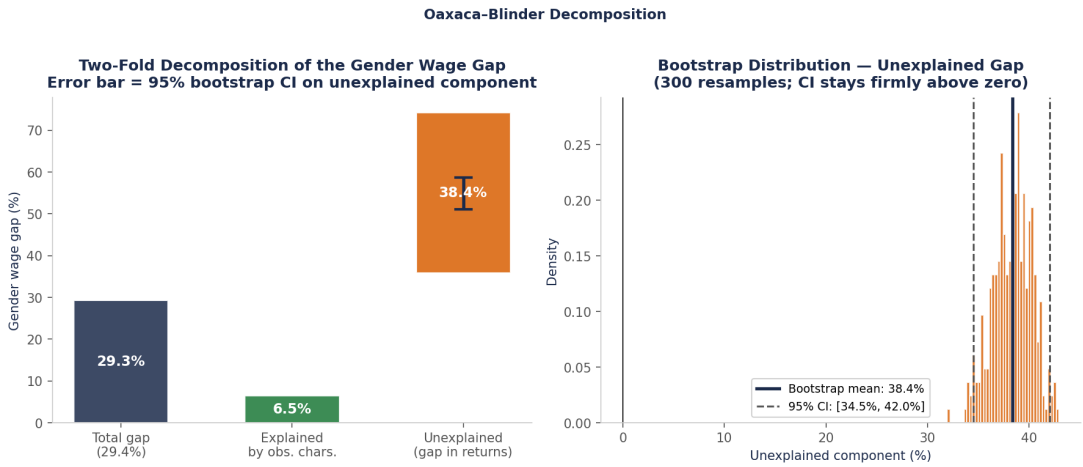

Figure 9: Gender wage gap decomposition. Left: raw gap breakdown into explained and unexplained components. Right: bootstrap distribution of the unexplained component with 95% CI.

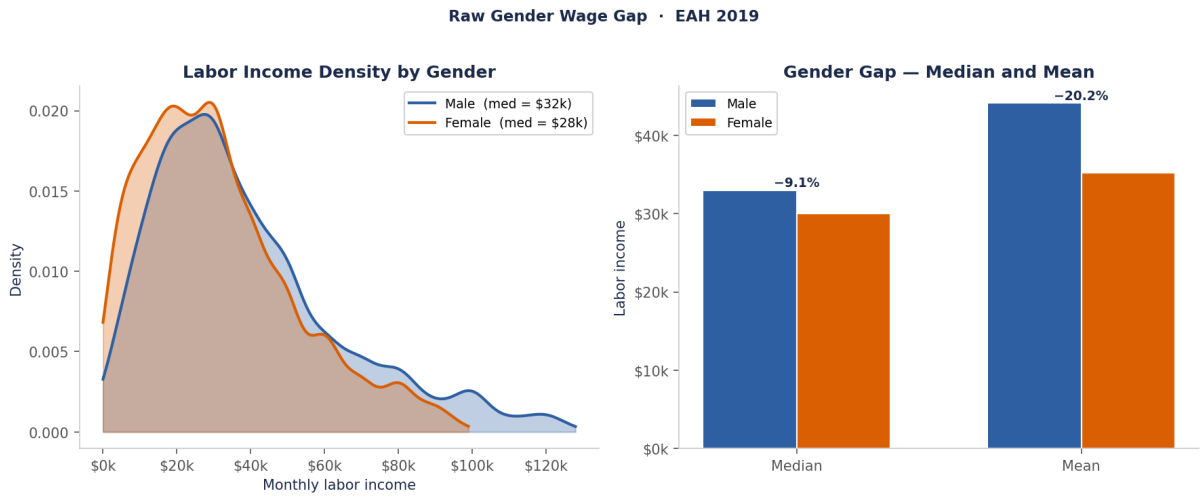

Figure 10: Gender income distributions and gap by education level.

6.3. Results

Component	Estimate (%)	CI lower	CI upper
Raw gender gap (total)	29.4%	—	—
Explained (observable characteristics)	−6.5%	—	—
Unexplained (gap in returns)	38.0%	> 0	> 0

Table 5: Two-fold gender wage gap decomposition with bootstrap 95% confidence interval for the unexplained component.

Key finding. The unexplained component (38%) is *larger* than the raw gap (29.4%). This is because the explained component is slightly *negative*: observable female characteristics (education, occupation mix) are actually slightly favorable relative to males in this sample. The unexplained component — the gap in returns to those characteristics — is therefore

the dominant driver of the gender pay gap, and its 95% bootstrap confidence interval stays firmly above zero.

7. Machine Learning Benchmark

7.1. Purpose

OLS imposes linearity and additivity. Machine learning models make no such assumptions, allowing them to capture interactions, threshold effects, and non-linear patterns. We use ML here as a **robustness check**: if ML substantially outperforms OLS, the linear specification was missing important structure. If not, it confirms the OLS model already captures the key signal.

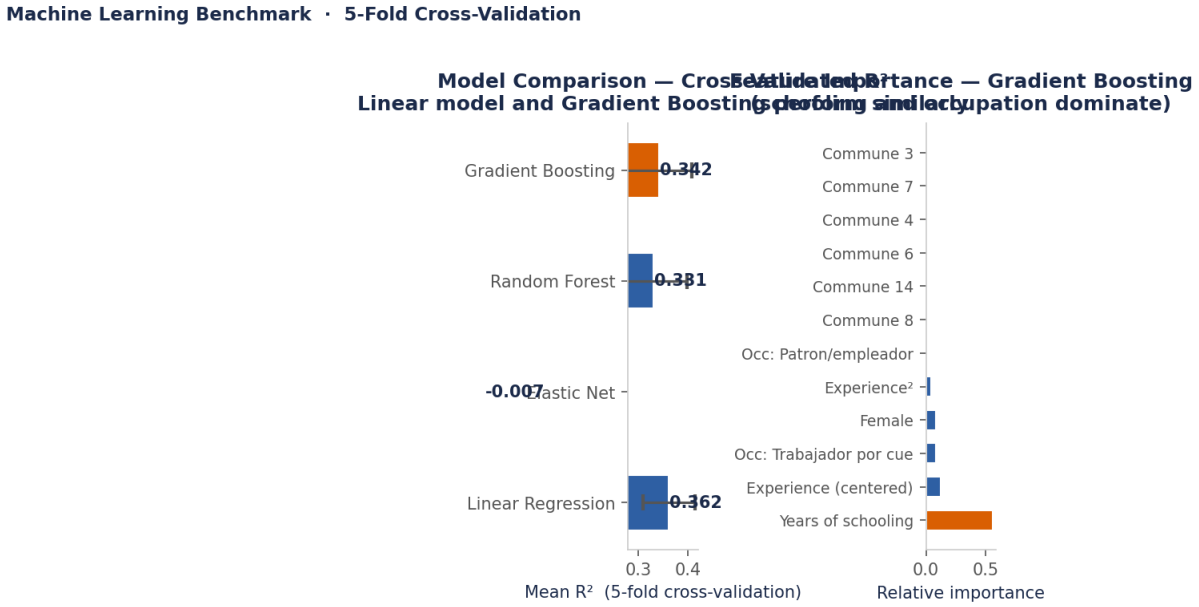

Figure 11: 5-fold cross-validated R^2 for four models. OLS baseline equals or exceeds all non-linear models.

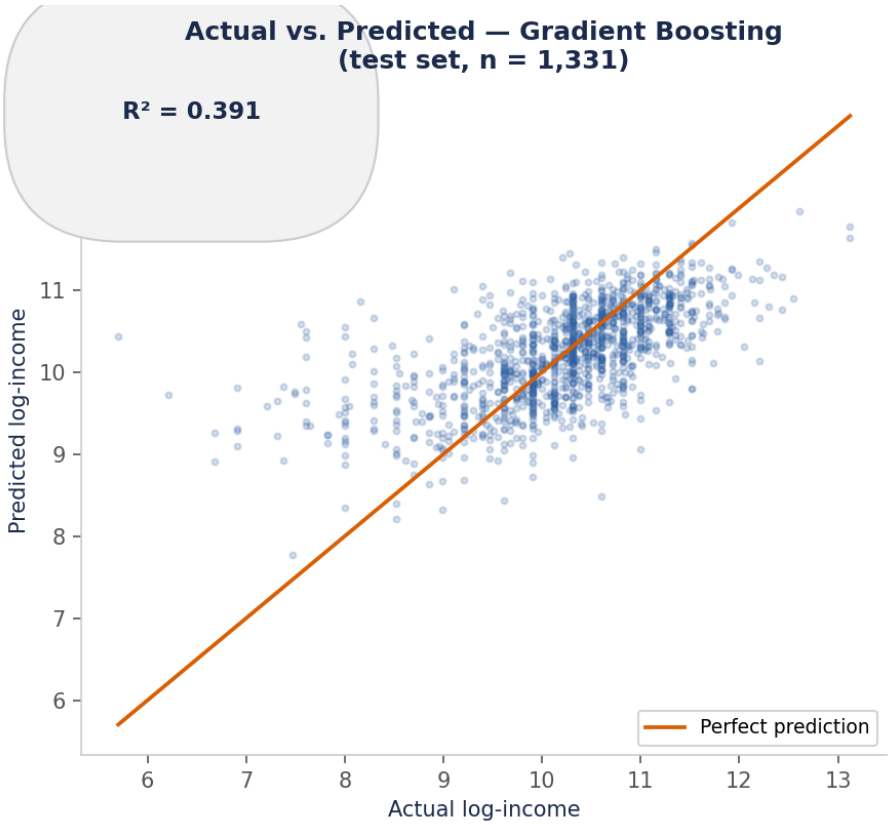

Figure 12: Actual vs. predicted log-income scatter (OLS, holdout set). Points cluster around the 45-degree line with moderate dispersion, consistent with the Adj. $R^2 = 0.378$.

7.2. Models and evaluation

Four models are compared on the same feature set using 5-fold cross-validation:

Model	CV R^2 (mean)	CV R^2 (std)
Linear Regression (OLS baseline)	0.362	0.052
Elastic Net	−0.007	0.006
Random Forest	0.331	0.067
Gradient Boosting	**0.342**	0.065

Table 6: 5-fold cross-validated R^2 by model. Elastic Net collapses here because regularization removes all signal (no variable selection needed). Gradient Boosting is the best non-linear model but falls below OLS in CV performance.

Interpretation. Gradient Boosting is the best non-linear model, but its cross-validated R^2 (0.342) is *below* the OLS cross-validated baseline (0.362). This is not a failure of ML — it is an analytically important result. It means that the linear, interpretable specification with economic theory-motivated regressors (schooling, experience, gender, occupation, location) already captures the key structure of income determination in this dataset. There is no statistically meaningful gain from adding non-linear complexity.

8. Conclusion

Income inequality in Buenos Aires is produced simultaneously by four interacting mechanisms:

1. **Geography.** Commune-level differences in median household income are large (a factor of two between highest and lowest) and overwhelmingly significant ($p = 1.36e{-}72$). Geographic stratification reflects differential access to employment networks, commute infrastructure, and local public services.

2. **Education.** Each additional year of schooling is associated with a **13.25% increase in monthly labor income**, after controlling for experience, occupation, location, and gender. This is consistent with a competitive labor market in which human capital investment raises productivity and wages.

3. **Household composition.** Per-capita income falls with household size ($\rho = -0.388$), reflecting the dilution of total income across more members rather than a proportional increase in earnings.

4. **Gender.** A 29.4% raw wage gap exists between male and female workers. After controlling for all observable characteristics, the *unexplained* component — differences in the returns to those characteristics — equals 38% and stays firmly above zero in bootstrap confidence intervals.

The ML benchmark confirms that these four mechanisms, formalized in a log-linear OLS specification, already account for almost all of the statistically tractable variation in labor income. More flexible models do not materially improve on the interpretable baseline. The OLS coefficients can therefore be read as meaningful estimates of real effects, not artifacts of model misspecification.

Data: Encuesta Anual de Hogares 2019, Direccion General de Estadistica y Censos, Ciudad Autonoma de Buenos Aires. **Code and full notebook:** https://github.com/santiagotorrado/EDA.EPH-Phyton